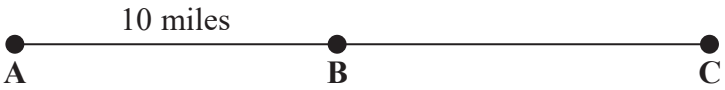
6 The diagram shows some information about junctions **A**, **B** and **C** on a motorway.



Raja drove from **A** to **B** at an average speed of 50 mph.
The distance from **A** to **B** is 10 miles.

Raja took 30 minutes to drive from **A** to **C**.
He drove from **A** to **C** at an average speed of 62 mph.

Work out Raja's average speed as he drove from **B** to **C**.

Whole journey: 30 minutes = 0.5 hours

AC = 62 × 0.5 = 31 miles

1 mark
Distance = speed × time.

AB time = 10 ÷ 50 = 0.2 hours = 12 minutes

1 mark
Time = distance ÷ speed.

BC distance = 31 – 10 = 21 miles

BC time = 30 – 12 = 18 minutes = 0.3 hours

Speed on BC = 21 ÷ 0.3 = 70 mph

1 mark
Use BC distance and time.

1 mark
70 mph.

..... mph

(Total for Question 6 is 4 marks)

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